MA 765: Topics in Non-linear Algebra

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1 Gröbner Basis

Denote by $\leq$ the coordinate wise partial order on $\mathbb{N}_0^n$. $(a_1, \dots, a_n) \leq (b_1, \dots, b_n)$ if $a_i \leq b_i$ for $i \in [n]$.
Divisibility is a partial order on monomials.

Theorem 1.1 (Dicksen's lemma)**.** Every infinite subset of $\mathbb{N}_0^n$ contains elements a, b with $a < b$.

Proof. The proof follows by induction. Let $\mathcal{M} \subset \mathbb{N}_0^n$ be an infinite subset. For any $i \in \mathbb{N}_0$ define $\mathcal{M}$,

$$\mathcal{M}_i = \{a = (a_1, \dots, a_n) \in \mathbb{N}_0^{n-1} : (a, i) \in \mathcal{M}\}$$

If $\mathcal{M}_i$ is finite, then look at $\bigcup_i \in \mathbb{N}_0$, which has finitely many and atleast one minimal element. Thus there is some $j \in \mathbb{N}$ such that $\bigcup_{i=0}^{j} \mathcal{M}_i$ with $a \in \mathcal{M}_i$ for some $i \leq j$.Hence $(a, i) < (b, k)$.

□

Corollary 1.2. For any $\phi \neq \mathcal{M} \subset \mathbb{N}_0^n$ the set of minimal elemnents wrt $<$ is nonempty and finite.

1.3. Monomial order on $\mathbb{N}_0^n$ is a relation $\prec$ such that

1. If $a \neq b$ then $\prec b$ or $b \prec a$.
2. If $A \prec b$ and $b \prec c$ then $a \prec c$.
3. $(0, \dots, 0) \prec a$ for any $a \in \mathbb{N}_0^n$.
4. $a \prec b$ then $a + c \prec b + c$ for any $c \in \mathbb{N}_0^n$.

First two conditions together imply that $\prec$ is a total order.

Remark 1.4.
1. If $a < b$, then $a \prec b$, i.e., any monomial order refines coordinate wise partial order.
2. Any monomial order on $\mathbb{N}_0^n$ gives total order on monomials in $k[X_n]$

Example 1.5. Lexicographic order: Left most non zero entry of $a - b$ is positive the $a >_{lex} b$.

degree lex order if either $|a| > |b|$ or $|a| = |b|$ and right most entry of $a - b$ is negative.

degree reverse lex order, same as above but right most entry of $a - b$ is negative.

Proposition 1.6. For any monomial order $<$ on $\mathbb{N}_0^n$ with $\phi \neq \mathcal{M} \subset \mathbb{N}_0^n$ has a unique minmial element.

Proof. Dicksen's lemma gives that $\mathcal{M}$ has a finite nom empty set of minmal elements wrt coordinate wise order. The minmal elet of thse wrt $<$ is the desired elemnt. □

1.7. Fix a monomial order $<$ on $k[X_n]$

1. The initial monomial $\mathrm{im}_<(F)$ of $f = \sum c_a x^a$ is the largest monomial wrt $<$ appearing in f with non zero coefficients.
2. The leading term $\mathrm{lt}_<(f) := c_a x^a$ with $x^a = \mathrm{im}(f)$

3. The initial ideal of an ideal I is $\mathrm{im}(I) = \langle \mathrm{im}(f) : f \in I \rangle$.

If G is a generating set for an ideal I then $\langle \mathrm{im}(g) : g \in G \rangle \subset \mathrm{im}(I)$ and this inclusion can be strict.

Proposition 1.8. Let $<$ be a monomial order on $k[X_n]$. Every ideal I has a finite subest $\mathcal{G}$ such that $\mathrm{im}(I) = \langle \mathrm{im}(g) : g \in \mathcal{G} \rangle$.
Any such $\mathcal{G}$ is called a Gröbner Basis of I wrt $<$.

Proof. The set of monomials in $\mathrm{im}(I)$ has a finite and nonempty subset of minmal elements wrt divisibility, say $m_1, \dots, m_s$. Thus $\mathrm{im}(I) = \langle m_1, \dots, m_s \rangle$. Every monomial in $\mathrm{im}(I)$ is the initial monomial of some $f \in I$. hence there exists $f_1 \dots, f_s \in I$ with $\mathrm{im}(f_i) = m_i$. Thus $\{f_i\}$ is a Gröbner basis. □

Theorem 1.9. If $\mathcal{G}$ is a Gröbner basis of I, then $I = \langle \mathcal{G} \rangle$

Note that Hilbert's basis theorem is a simple corollary of this.

Proof. We argue by contradiction. By 1.6, choose $f \in I - \langle \mathcal{G} \rangle$ such that $\mathrm{im}(f)$ is minimal. Call $\mathrm{im}(f) = x^b$. $x^b \in \mathrm{im}(I) = \langle \mathrm{im}(g) : g \in \mathcal{G} \rangle$. There exists $g \in \mathcal{G}$ such that $\mathrm{im}(g) | x^b$, say $x^b = x^c \cdot \mathrm{im}(g)$.
$\mathrm{im}(f = x^c \lambda g) < x^b = \mathrm{im}(f)$ where $\lambda = \mathrm{lt}(f)/x^c \, \mathrm{lt}(g) \in k$. But $f - x^c \lambda g \in I - \langle \mathcal{G} \rangle$, which is a contradiction to the minimality. □

Lemma 1.10. Consider $f, g_1, \dots, g_s \in k[X_n]$, with $g_i \neq 0$. Then for any minmal order $<$, there exists $q_1 \dots, q_s, r \in k[X_n]$ such that

1. $f + \sum_{i=1}^{s} q_i g_i + r$
2. $\mathrm{im}(f) \geq \mathrm{im}(q_i g_i) \forall i$ (note that $\mathrm{im}(f) = \mathrm{im}(q_i g_i)$ for some i)
3. $\mathrm{im}(r)$ is not divisible by $\mathrm{im}(g_i)$ for any i.

We say f reduces to r by $\{g_1, \dots, g_s\}$.

1.11 Division Algorithm. Input: $f, g_1 \dots g_s$
Output: $q_1, \dots q_s, r$ satisfying the properties 1-3.

1. Set $r = 0$, $p = gf, q_1 = \dots = q_s = 0$
2. While $p \neq 0$ do :

   If $\mathrm{im}(g_i)$ divides $\mathrm{im}(p)$ for some $i \in [s]$, then set $q_i = q_i + \frac{lt(p)}{lt(g_i)}$ and $p_i = p - \frac{lt(p)}{lt(g_i)} g_i$
   else set $r = r + \mathrm{lt}(p)$, $p = p - \mathrm{lt}(p)$
3. Return $q_1, \dots, q_s, r$

Example 1.12. Consider Lexicographic ordering with $x > y$ and $f = x^2y + xy^2 + y^2$ and

$g_1 = xy - 1$, $g_2 = y^2 - 1$.

p	$xy - 1$	$y^2 - 1$	r
$x^2y + xy^2 + y^2 - x^2y + x$	x		
$xy^2 + y^2 + x - xy^2 + y$	y		
$y^2 + x + y - x$			x
$y^2 + y - y^2 + 1$	1		
$y + 1 - y$			y
1			1

Corollary 1.13. Buchberger's criteria Let $\mathcal{G}$ be a finite subset of I. Then $\mathcal{G}$ is a Gröbner basis of I iff each $f \in I$ can be reduced to 0 by $\mathcal{G}$.

Proof. If f reduces to r by $\mathcal{G}$, then $\text{im}(g_i)$ does not divide $\text{im}(r)$, for all i. Howevert $f - r \in \langle \mathcal{G} \rangle$ and $r \in I$ and $\mathcal{G}$ is a Gröbner basis of I. So there exists some $g \in \mathcal{G}$ such that $\text{im}(r)$ is divisible by $\text{im}(g)$, which forces $r = 0$.

Conversely, we have $f = \sum q_i g_i$ with $g_i \in \mathcal{G}$ and $\text{im}(q_i g_i) \leq \text{im}(f)$. Hence equality for some i and so $\text{im}(f) \in \langle \text{im}(g) : g \in \mathcal{G} \rangle$. □

Proposition 1.14. Let $<$ be a monomial order. Then

1. Let B be the set of monomials in $k[X_n] - \text{im}(I)$. Then $\bar{B} \subset k[X_n]/I$ is a k vsp basis.
2. If $\mathcal{G}$ is a Gröbner basis of I, then the remainder of f by $\mathcal{G}$ is unique and does not depnd on the choice of $\mathcal{G}$.

Proof.

1. If $p = \sum \lambda_i m_i \in I$ with $m_i \in B$, then $\text{im}(p) \in \text{im}(I)$, but $\text{im}(p) = \text{im}(m_i) \notin I$. To show $\bar{B}$ spans.$m \in k[X_n]$ such that $\bar{m} \notin$span$(\bar{B})$.

   Take $\min\{m\} = m$ where $m \notin B$. So we have $m \in \text{im}(I)$. There exists $f \in I$ such that $\text{im}(f) = m$. So any monomial in $f - \text{lt}(f) + I = f - \lambda m + I$ is in span $(\bar{B})$. So $\lambda m + I = p - f + I \in$span$(\bar{B})$. This leads to a contradiction

□

Definition 1.15.

1. For terms $\lambda x^a, \mu x^b \quad (\lambda, \mu \in k)$ denote by

$$\gcd\left(\lambda x^a, \mu x^b\right) = \gcd\left(x^a, x^b\right)$$
$$\text{lcm}\left(\lambda x^a, \mu x^b\right) = \text{lcm}\left(x^a, x^b\right)$$

2. For $0 \neq g$, h $\in k\left[x_n\right]$, their s -polynomial (wrt monomial order $<$)

$$S(g,h) := \frac{\text{lt}(h)}{\gcd(\text{lt}(h), \text{lt}(g))} g = \frac{\text{lt}(g)}{\gcd(\text{lt}(h), \text{lt}(g))} h.$$

1.16. Buchberger's Algorithm for computing Gröbner basis Input: $f_1, \ldots, f_s \in k\left[X_n\right]$ in monomial order.
Output: Gröbner basis $\mathcal{G}$ of $I = \langle f_1, \ldots, f_s \rangle$ wrt $<$.

1. Set $\eta = \langle f_1 \cdots, f_s \rangle$
2. Order the elements of $\mathcal{G}$ as $f_1, \dots, f_t$
3. For $1 \leq i < j \leq t$ do: Reduce $S(g_i, g_j)$ to r by g. If $r \neq 0$, then set $\mathcal{G} := \mathcal{G} \cup \{r\}$ and go to step 2 .
4. Return $\mathcal{G}$

Remark 1.17. The algorithm computes a Gröbner basis. It terminate vecause in case $r \neq 0.\, \mathrm{im}(r) \notin \langle \mathrm{im}\,(g_1)\,, \cdots, \mathrm{im}\,(g_t) \rangle$.

1.18. Extension to submodules of finitely generated $k[X_n]$ module So $F = k[X_n]^r = \bigoplus_{i=1}^{r} k[X_n]e_i$.

Monomials in G are of the form $x^a e_i$ and terms are $\lambda x^a e_i$ with $\lambda \in k$.
A monomial order on F is a total order on the monomials satisfying:
If $x^a \neq 1$, then $m_1 < m_2 \implies m_1 < x^a,_1 < x^a m_2$, for monomials m_i in F.

Given a monomial order on the polynomial ring $k[X_n]$ and an order on $\{e_i\}$. We obtain a monomial ordering on F by ordering $\mathbb{N}_0^n \times [r]$ or $[r] \times \mathbb{N}_0^r$ lexicographically.

Dicksen's lemma can be extended to monomials in F. We also define im, lt with respect to $<$ analogously. There is also a division algorithm.

Theorem 1.19.
1. Every submodule M of F has finite Gröbner basis and the basis generates M.
2. If B is the set of monomials in $F - \mathrm{im}(M)$, then $\bar{B} \subset F/M$ form a k basis of F/M.

2 Hilbert Functions

Definition 2.1. Let $I \subset k[X_n]$ be a monomial ideal . The Hilbert function of $A = k[X_n]/I$ (or of I) is

$$\begin{aligned} h_A \ : & \mathbb{N}_0 \to \mathbb{Z} \\ & a \mapsto h_A(j) = \dim_k [A]_j \end{aligned}$$

where $[A]_j$ is k vector space of images of polynomials of degree j, from $k[X_n] \to A$.(So $h_A(j)$=number of monomials in $[k[X_n]]_j - I$).

It's generating function is the Hilbert series

$$H_A(z) = \sum_{j \geq 0} h_A(j) z^j$$

Example 2.2. 1. For $I = 0$, we get $h_{k[X_n]}(j) = \binom{n+j-1}{j}$ and so

$$H_{k[X_n]}(z) = \sum_{j \geq 0} \binom{n+j-1}{n-1} z^j = \frac{1}{(1-z)^n}$$

2. If $I = \langle x^a \rangle$, then let $e = \deg(x^a)$.

3.

$$h_{k[X_n]/I}(j) = \begin{cases} h_{k[X_n]}(j) & j < e \\ h_{k[X_n]}(j) - h_{k[X_n]}(j-e) & j \geq e \end{cases}$$

Hence

$$H_{k[X_n]/I} = \frac{1 - z^e}{(1-z)^n}$$

Theorem 2.3. For any proper monomial ideal I of $k[X_n]$, the Hilbert series of $A = k[X_n]/I$ is a rational function of the form

$$H_A(z) = \frac{\kappa_A(z)}{(1-z)^d}$$

with $\kappa_A(z) \in \mathbb{Z}[z], \kappa_A(0) = 1, \kappa_A(1) \neq 0, d \in \mathbb{N}$.(this expression is unique.)

The dimension of A is

$$\dim A \ := d$$

and the multiplicity of A (or degree of I) is

$$\deg(I) = \mathcal{H}_A(1) > 0$$

There is a polynomial called Hilbert polynomial p_A of A such that $h_A(j) = p_A(j)$ if $j >> 0$. If $d = \dim A > 0$, then

$$\begin{aligned} p_A(z) = & \frac{\deg(I)}{(d-1)!} z^{d-1} + \text{lower order terms} \\ = & h_0(A) \binom{z+d-1}{d-1} + h_1(A) \binom{z+d-2}{d-2} + \cdots + h_{d-1}(A) \binom{z}{0} \end{aligned}$$

where $h_0(A) = \deg(I)$ and $h_i(A)$ are integers. Note if $p_A(z) \neq 0$ polynomial, then $\deg p_A = d - 1 = \dim A - 1$.

Example 2.4.

$I = 0$, we get $\dim k[X_n] = n$ and the multiplicity of $k[X_n]$ is $1 = \deg I$.

$I = \langle x^a \rangle$ with $\deg x = e$, then $\dim k[X_n]/I = n - 1$ and $\deg(I) = \deg(x^a)$.

Proof of 2.3. Inclusion exclusion principle: $| \bigcup_{i=1}^{s} X_i | = \sum\limits_{\phi \neq T \subset [S]} (-1)^{|T|+1} \mid X_t \mid$ where $X_t = \bigcap_{i \in T} X_i$

Let $I = \langle m_1, \dots, m_s \rangle = I$, where m_i are monomials. Denote by $X_i(j)$ set of degree j monomials in $\langle m_i \rangle \subset k[X_n]$. hence for $T \subset [s]$, $X_T(j) = \bigcap_{i \in T} X_i(j)$ is the set of deg j monomials that are divisible by $m_T = \operatorname{lcm}(m_i)_{i \in T}$. Define $e_t := \deg m_T$. So

$$|X_T(j)| = \begin{cases} 0 & j < e_t \\ \binom{n-1+j-e_T}{n-1} & j \geq e_T \end{cases}$$

Thus

$$\sum_{j \geq 0} |X_T(j)| z^j = \sum_{j \geq e_T} \binom{n-1+j-e_T}{n-1} z^j = \sum_{k \geq 0} \binom{n-1+k}{n-1} z^{k+e_T} = z^{e_T} \frac{1}{(1-z)^n}$$

Since

$$\begin{aligned} h_A(j) &= \binom{n-1+j}{n-1} - \text{number of deg } j \text{ monomials in } I = \langle m_1 \dots, m_s \rangle \\ &= \binom{n-1+j}{n-1} - | \bigcup_{i \in [j]} X_i | \end{aligned}$$

We get

$$\begin{aligned} H_A(z) = \sum_{j \geq 0} h_A(j) z^j &= \sum_{j \geq 0} \binom{n-1+j}{n-1} z^j + \sum_{T \in [s]} (-1)^{|T|} \sum_{j \geq 0} |X_T(j)| z^j \\ &= \frac{1}{(1-z)^n} + \sum_{T \in [s]} (-1)^T \frac{z^{e_T}}{(1-z)^n} =: \frac{g(z)}{(1-z)^n} \end{aligned}$$

When $g(z) \in \mathbb{Z}[z]$, with $g(0) = 1$ writing $g(z) = (1-z)^v \kappa_A(z)$ with $\kappa_A(z) \in \mathbb{Z}[z]$, suitable $v \in \mathbb{N}$ and $\kappa_A(1) \neq 0, \kappa_A(0) = 1$. Hence $H_A(z) = \frac{\kappa_A(z)}{(1-z)^d}$ where $d = n - v$. Write

$$\kappa_A(z) = \sum_{k=0}^{w} c_k z^k \text{ with } c_k \in \mathbb{Z}$$

$$\sum_{j \geq 0} h_A(j) z^j = \Big(\sum_{k=0}^{w} c_k z^k \Big) \Big(\sum_{l \geq 0} \binom{d-1+l}{d-1} z^l \Big)$$

comparing coefficients in deg $j >> 0$, we get

$$h_A(j) = \sum_{k+l=j} c_k \binom{d-1+l}{d-1} = \sum_{k=0}^{w} c_k \underbrace{\binom{d-1+j-k}{d-1}}_{\substack{\text{polynomial in } j-k \\ \text{variables of deg } d-1}}$$

$$= \underbrace{\left(\sum_{k=0}^{w} c_k\right)}_{\kappa_A(1)} \binom{j}{d-1} + \text{lower order terms}$$

$$=: \ p_A(j) \text{ (Hilbert polynomial)}$$

If $h_A(j) = 0$ whenever $j >> 0$, then by definition $0 = d = \dim A$ and in this case p_A is the zero polynomial. Hence if $d > 0$, then $h_A(j) > 0$ if $j >> 0$ and so the leading coefficient of $p_A(z)$ must be positive, i.e., $\kappa_A(1) > 0$ $\square$

Definition 2.5. A monomial order is called degree compatible if

$$\deg(x^a) > \deg(x^b) \implies x^a > x^b$$

for any two monomials.

2.6. For any ideal $I \subset k[X_n]$ and any $t \in \mathbb{Z}$, set

$$I_{\leq t} = \{f \in I \ : \ \deg f \leq t\}$$

It is a k-subspace of $k[X_n]$. Write $\mathrm{Mon}(k[X_n])$ for the set of monomials in $k[X_n]$.

Lemma 2.7. Let $<$ be a degree compatible monomial order. For any ideal $I \subset k[X_n] = S$, one has

$$\dim_k \frac{k[X_n]_{\leq t}}{I_{\leq} t} = \text{number of monomials in } \frac{k[X_n]_{\leq t}}{\mathrm{im}_<(I)}$$
$$= \mid \mathrm{Mon}(k[X_n])_{\leq t} - \mathrm{im}_<(t) \mid$$

Proof. $B := \mathrm{Mon}(k[X_n])_{\leq t} - \mathrm{im}_<(I)$. We claim

$$\bar{B} \subset \frac{k[X_n]_{\leq t}}{I_{\leq t}} \text{ is a } k\text{- basis}$$

$\overline{\mathrm{Mon}(k[X_n]) - \mathrm{im}(I)}$ is a k-basis of $k[X_n] - \mathrm{im}(I)$, by 1.14. $\bar{B}$ spans remainder of any $F \in k[X_n]$ pon dividing by Gröbner basis of I wrt $<$ satisfies $\deg r \leq \deg f$. $\square$

For $I \subset k[X_n]$, the affine Hilbert funtion of $A = k[X_n]/I$ is

$$h_A^a : \mathbb{N}_0 \longrightarrow \mathbb{Z}$$
$$j \mapsto h_A^a(j) = \dim_k \frac{k[X_n]_{\leq j}}{I_{\leq j}}$$

Lemma 2.8. For any degree compatible monoial order $<$ on I, one has

$$h_{\frac{k[X_n]}{\mathrm{im}_<(I)}}(j) = h_A^a(j) - h_A^1(j-1)$$

where $A = k[X_n]/I$.

Proof. By definition, we have

$$h_{\frac{k[X_n]}{\mathrm{im}(I)}}(j) = \mid \mathrm{Mon}(k[X_n])_j - \mathrm{im}(I) \mid$$

$$2.7 \Longrightarrow \; h^a_A(j) = \mid \mathrm{Mon}(k[X_n])_{\leq j} - \mathrm{im}(I) \mid$$ □

Remark 2.9. If $A \neq 0$, $h^a_A(0) = 1$. Then by 2.8 we have $h^a_A(j) = \sum_{k=0}^{j} h_{k[X_n]/\mathrm{im}(I)}(k)$. It follows that generating function of h^a_A and $h_{\frac{k[X_n]}{\mathrm{im}_<(I)}}$ have analogous properties

2.10. We define dimensin of A,

$$\dim \frac{k[X_n]}{I} := \dim_k \frac{k[X_n]}{\mathrm{im}(I)}$$

and the degree

$$\deg(I) = \deg(\mathrm{im}(I))$$

where $<$ is a degree compatible monomial order.

2.11. Let $G = (G, +)$ be an abelian group. A G-graded ring R is a family of subgroups $([R]_{a \in G}) \leq (R, +)$ such that

1. $R = \oplus_{a \in G}[R]_a$ (as $\mathbb{Z}$-modules)
2. $[R]_a \cdot [R]_b \subset [R]_{a+b}$

The elements of $[R]_a$ are called homogeneous of degree a.

Example 2.12. Fine or $\mathbb{Z}$-grading of $k[X_n]$, where $G = \mathbb{Z}^n$

$$[S]_a = \begin{cases} 0 & \text{if some } a_i < 0 \\ \{\lambda x^a \ : \ \lambda \in k\} & \end{cases}$$

Definition 2.13. $R = G-$graded ring

1. G-graded R-module is an R-module M with a decomposition $([M]_a)_{a \in G}$ such that $M = \oplus_{a \in G}[M]_a$ and $[R]_a \cdot [M]_b \subset [M]_{a+b}$.
2. A G-graded or simply graded submoudle of such a graded M is a graded submodule $N \subset M$ such that
$$[N]_a \subset [M]_a$$

Lemma 2.14. For an arbitrary submodule N of a G graded R-module M the following are equivalent

1. N is a graded submodule
2. N has a generating set consisting of homogeneous elemnents
3. If $m = \sum_{a \in G} m_a$ with $m_a \in [M]_a$, then $m \in N$ iff each $m_a \in N$.
4. M/N is a G-graded R-module with grading $[M/N]_a := \frac{[M]_a + N}{N}$.

Proof. □

Example 2.15. 1. M and N are G-graded, then so is $M \oplus N$ with grading $[M \oplus N]_a := [M]_a \oplus [N]_a$ (as R modules). So if R is G graded, then so is R^n.

2. $I \subset k[X_n]$ is a $\mathbb{Z}^n$ graded submodule if I is a monomial ideal.

3. A $\mathbb{Z}$-graded or homogeneous ideal of $k[X_n]$, is an ideal that has generating set consisting of homogeneous polyonomials. In that case $k[X_n]/I$ is a graded module.

2.16. A homomorphism of G-graded modules or a G-graded homomorphism is a R-module homomorphism $\phi : M \to N$ that is degree preserving, $\phi([M]_a) \subset [N]_a$.

For any $a \in G$ and a G-graded module M, the module $M(a)$ has the same module structure as M, but grading given by

$$[M(a)]_b := [M]_{a+b}$$

$M(a)$ is a degree a shift of M. (Note here that the convention is opposite of that in algebraic topology.)

Example 2.17. 1. Consider $k[X_n]$ with standard grading.

$$\begin{aligned} \phi : k[X_n] &\to k[X_n] \\ f &\mapsto x_1^2 f \end{aligned}$$

is not a graded homomorphism. However define

$$\begin{aligned} \psi : k[X_n](-2) &\to k[X_n] \\ f &\mapsto x_1^2 f \end{aligned}$$

then $f \in k[X_n](-2)$ has degree $\deg f + 2$. So $f \in [k[X_n](-2)]_{\deg f+2}$.

2. For any $a \neq$ in G, $R(a)$ is not a graded ring, (because identity is not in 0 dimension), but it is a graded R-module.

Lemma 2.18. If $\phi : M \to N$ is a homomorphism of graded modules, then $\ker \phi, \mathrm{im}\phi, \mathrm{coker}\phi$ are graded modules.

Proof. $[\ker \phi]_a = \ker \phi \bigcap [M]_a$, and $[\mathrm{im}\phi]_a = \mathrm{im}\phi \bigcap [N]_a$.

□

Example 2.19. 1. If M is a $\mathbb{Z}$ graded module with generators $m_1, \dots, m_t$ where $d_i = \deg m_i$, then

$$\begin{aligned} \phi : \bigoplus_{i=1}^{t} R(-d_i) &\longrightarrow M \\ \begin{bmatrix} r_1 \\ \vdots \\ r_t \end{bmatrix} &\mapsto \sum_{i=1}^{t} r_i m_i \end{aligned}$$

is a homomorphism of graded R-modules and is surjective.

2. Consider $I = \langle x^3, xy, y^4 \rangle \subset k[x, y] + S$ with standard grading.

$$\phi : S(-3) \oplus S(-2) \oplus S(-4) \to I$$
$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \mapsto f_1x^3 + f_xy + f_3y^4$$

$$\ker\phi : \left\langle \begin{bmatrix} y \\ -x^2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ y^3 \\ -x \end{bmatrix} \right\rangle \xleftarrow[\text{graded hom}]{\simeq} S(-4) \oplus S(-5)$$

There exists an exact sequence,

$$0 \to \begin{matrix} S(-4) \\ \oplus \\ S(-5) \end{matrix} \xrightarrow[\begin{bmatrix} y & 0 \\ -x^2 & y^3 \\ 0 & -x \end{bmatrix}]{} \begin{matrix} S(-3) \\ \oplus \\ S(-2) \\ \oplus \\ S(-4) \end{matrix} \xrightarrow[\begin{bmatrix} x^3 & xy & y^4 \end{bmatrix}]{\phi} S \to S/I \to 0$$

Definition 2.20. For any $\mathbb{Z}$ graded module M over $k[X_n]$ its Hilbert function is

$$h_m : \mathbb{Z} \to \mathbb{Z}$$
$$j \mapsto h_m(j) := \dim_k[M]_j$$

assuming $[M]_j$ is finitely generated for all j.

Remark 2.21. For any monomial ideal $I \subset k[X_n]$,2.1 and 2.20 agree.

$$\dim_k[k[X_n]/I]_j =| [\mathrm{Mon}(k[X_n])]_j - I |$$

Proposition 2.22. For every graded submodule M of a finitely generated free $k[X_n]$ module F and any monomial order $<$ of F,

$$h_{F/M}(j) = \frac{h_F(j)}{\mathrm{im}(M)} \ \forall j \in Z$$

Corollary 2.23. For any finitely generated $k[X_n]$ submodule $m \neq 0$, Hilbert series is of the form

$$H_M(z) := \sum_{j \in Z} h_M(j) z^j$$
$$= \frac{\kappa_M(z) z^t}{(1-z)^d}$$

where $\kappa_M(z) \in \mathbb{Z}[z]$, $\kappa_M(0) \neq 0, \kappa_M(1) > 0, t \in \mathbb{Z}$.

There is a HIlbert polynomial $p_M(z) \in \mathbb{Q}(z)$ such that

$$h_M(j) = p_M(j) \ j >> 0$$

(krull) dimension of M is defined as

$$\dim M = d$$

and the degree is

$$\deg(M) = \kappa_M(1)$$

Proof. Let M be generated by $m_1, \dots, m_t$ of degree $d_1, \dots, d_t$ respectively. Define

$$\begin{aligned} \phi \, : \, \oplus_{i=1}^{t} S(-d_i) &\twoheadrightarrow M \\ f &\mapsto x_1^2 f \end{aligned}$$

is not a graded homomorphism. However define

$$\begin{aligned} \psi \, : \, k[X_n](-2) &\to k[X_n] \\ \begin{bmatrix} f_1 \\ vdots \\ f_t \end{bmatrix} &\mapsto \sum f_i m_i \end{aligned}$$

Set $N = \ker \phi$ and we have a short exct sequence $0 \to N \to F \to M \to 0$ and by rank nullity theorem we have $h_M(j) = h_F(j) - h_N(j)$. S and F have desired Hilbert series which are rational functions. So it is enough to show the same for N, which is same as $h_{F/\operatorname{im}(N)}$.

$\operatorname{im}(N)$ is generated by monomials. So

$$\frac{F}{\operatorname{im}(N)} \cong \bigoplus_{i=1}^{t} \underbrace{\left(\frac{S}{J_i}\right)}_{\substack{\text{has the} \\ \text{desired} \\ \text{properties}}} (-d_i)$$

for monomial ideals J_i. $\square$

Example 2.24. 1. $S = k[X_n], F - \bigoplus_{i=1}^{t} S(-d_i)$. Then

$$H_{S(-d_i)}(z) = \frac{z_i^d}{(1-z)^n}$$

2. Consider $I = \langle x^3, xy, y^4 \rangle \subset k[x,y] + S$ with standard grading. $A = S/I$. We have the exact sequence,

$$0 \to \begin{matrix} S(-4) \\ \oplus \\ S(-5) \end{matrix} \to \begin{matrix} S(-3) \\ \oplus \\ S(-2) \\ \oplus \\ S(-4) \end{matrix} \xrightarrow[\begin{bmatrix} x^3 & xy & y^4 \end{bmatrix}]{\phi} S \to A = S/I \to 0$$

$$\begin{aligned} H_A(z) =& H_S(z) - H_{S(-3)\oplus S(-2)\oplus S(-4)}(z) + H_{S(-4)\oplus S(-5)}(z) \\ =& H_S(z) - H_{S(-3)}(z) - H_{S(-2)}(z) + H_{S(-4)}(z) + H_{S(-5)}(z) - H_{S(-4)}(z) \\ =& \frac{1 - z^3 - z^2 + z^5}{(1-z)^2} \\ =& 1 + 3z + 2z^2 + z^3 \end{aligned}$$

So $\dim A = 0$ and $\deg M = 6$ (polyonomial evaluated at 1).

Lemma 2.25. Let $d \in N$. Every $a \in \mathbb{N}$ admits a unique presentation of the form

$$a = \binom{k_d}{d} + \binom{k_{d-1}}{d-1} + \cdots + \binom{k_s}{s}$$

with integers $k_d > k_{d-1} \cdots > k_s$. It is called the d-Macaulay presentation of a.

Example 2.26. For $d = 3$ and $a = 12$, we have

$$12 = \binom{5}{3} + \binom{2}{2} + \binom{1}{1}$$

Definition 2.27. If $a > 0$ with d-Macaulay presentation

$$a = \binom{k_d}{d} + \binom{k_{d-1}}{d-1} + \cdots + \binom{k_s}{s}$$

set

$$a^{\langle d \rangle} = \binom{k_{d+1}}{d+1} + \binom{k_d}{d} + \cdots + \binom{k_{s+1}}{s+1}$$

Example 2.28. $12^{\langle 3 \rangle} = 17$

Theorem 2.29 (Macaulay). Let $h : \mathbb{N}_0 \to \mathbb{Z}$, the following are equivalent,

1. There is some $n \in \mathbb{N}_0$ and some homogeneous ideal $I \subset k[X_n]$ such that Hilbert function of $A = k[X_n]/I$ is h.
2. There is a monomial ideal (Lexicographic ideal) $I \subset k[X_n]$ with $n = h(1)$ such that Hilbert function of A is h.
3. $h(0) = 1$ and
$$h(j+1) \leq h(j)^{\langle j \rangle} \text{ if } j > 0$$

   Moreover for every graded k-algebra A, one has

$$h_A(j+1) = h_A(j)^{\langle j \rangle} \text{ if } j >> 0$$

Example 2.30.

j	0	1	2	3	4	5	6
$\tilde{h}$	1	4	10	12	18	18	$\cdots$
h	1	4	10	12	17	17	$\cdots$

$\tilde{h}$ is not a possible Hilbert function because $12^{\langle 3 \rangle} = 17$. While h is a possible Hilbert function.

3 Ideals and Schemes

3.1 Affine case

Definition 3.1. A ring R is reduced if $r^n = 0$ for some $n \in \mathbb{N}$ implies $r = 0$.

Lemma 3.2. Considere a reduced ring R and an ideal I of R. Then R/I is reduced iff $I = \sqrt{I}$.

Hilbert's Nullstellensatz gives bijection if $k = \bar{k}$. Let $S = k[x_1, \dots, x_n]$.

$$\left\{\begin{matrix}\text{Subvarieties of}\\ \mathbb{A}^n_k\end{matrix}\right\} \underset{Z}{\overset{I}{\leftrightarrows}} \left\{\begin{matrix}\text{Radical Ideals of}\\ S\end{matrix}\right\} \leftrightarrows \left\{\begin{matrix}\text{Reduced factor}\\ \text{rings of } S\end{matrix}\right\}$$

$$J \mapsto S/J$$

$$\ker(S \to A) \leftarrow A$$

Definition 3.3. The geometric object X associated to an ideal $J \subset S$ is called an affine (sub) scheme of $\mathbb{A}^n_k$. $I_X := J$ is called the defining ideal of X and S/J is called co-ordinate ring. $X = \operatorname{spec}(S/J)$ to denote the scheme X. The reduced subscheme of X is $X_{red} = \operatorname{spec}(S/\sqrt{J})$. It is also called the *support* of X.

Remark 3.4.
1. Definition 3.3 is a special case of an affine scheme. $\operatorname{Spec}(S/J)$ is the set of prime ideals of S/J endowed with Zariski topology where closed sets are of the form $V(p)$ where p is a prime in S containing J.
2. If $k = \bar{k}$, then the points of $X = \operatorname{Spec}(k[X_n]/\sqrt{J}) \subset \mathbb{A}^n_k$ are the points of $X_{red} = Z(J) = Z(\sqrt{J})$. (The scheme X captures more information, for example multiplicities of the common zeroes)

Example 3.5. For any $j \in \mathbb{N}$ the scheme $Y_j \subset \mathbb{A}^n$ defined by $(x_1, \dots, x_n)^j$ is supported at the point $(0, \dots, 0)$ i.e. $(Y_j)_{red} = \{(0, \dots, 0)\}$. Sometimes Y_j is called a *fat point*.

Definition 3.6. The dimension of $Y = \operatorname{Spec}(k[X_n]/J)$ is $\dim Y = \dim k[X_n]/J$ (as defined using Hilbert series 2.3)

Example 3.7. $\dim \operatorname{Spec} \frac{k[X_n]}{(x_1,\dots,x_n)^j} = 0 \quad \forall j$

Definition 3.8. Let $X, Y \subset \mathbb{A}^n$ be subschemes of $\mathbb{A}^n$. Then X is called a subscheme of Y, if $I_Y \subset I_X$. In symbols $X \subset Y$.

The intersection $X \cap Y$ is the scheme defined by $I_X + I_Y$ and the union $X \cup Y$ is defined by $I_X \cap I_Y$.

Example 3.9. Continuing with the notation used in 3.5, we have $Y_1 \subsetneq Y_2 \subsetneq \cdots$, but $Y_2 \not\subseteq \underbrace{\operatorname{Spec}\left(\frac{k[X_n]}{(x_2, \dots, x_n)}\right)}_{\text{line}}$ ("fat point sticks out of line")

Theorem 3.10. Primary decomposition theorem

Example 3.11. An affine scheme $Y \subset \mathbb{A}^n$ is irreducible if for any subschemes $Y_1, Y_2 \subset Y$ with $Y_1 \cup Y_2 = Y$ either $Y = Y_1$ or $Y = Y_2$ or $Y_{red} = (Y_1)_{red} = (Y_2)_{red}$

Lemma 3.12. 1. Y is irreducible iff I_Y is primary.

2. Y is irreducible and reduced iff I_Y is a primary ideal.

3.13. 1. The fat points Y_j in 3.5 are irreducible but not reduced if $j \geq 2$

2. A line Y is defined by $I_Y = \langle l_1, \dots, l_{n-1} \rangle$, where l_i are linear independent polynomials in $k[X_n]$. Any line is reduced and irreducible.